ACTSC 372 Investment Science and Corporate Finance

University of Waterloo - Fall 2025

24th October 2025

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Contents

0 The Role of Financial Intermediaries

Lecture 1, 2025/09/03

What is financial investment?

- Ownership / loan contracts helped by financial intermediaries.

Role of Financial Intermediaries

- **Size intermediation**: the bank gets small lenders together with the “big” borrowers.
- **Term intermediation**: between short term lenders and long term borrowers.
- **Risk intermediation**: pooling resources reduces individual risk.

Consumer Smoothing: see handwritten notes.

1 Investment Rules

1.1 The NPV Rule

Question: Given a choice of projects, what project should a rational investor choose?

Definition 1.1 (**Project**).

A **project** is a collection of cash flows $\{C_0, C_1, C_2, ...\}$ where C_t = cash flow at time t.

Note.

- $C_t < 0 \implies$ cost.
- $C_t > 0 \implies$ revenue.

Definition 1.2 (**Pure Investment Project**).

A **pure investment project** is a project where $C_0 < 0$ and $C_t \geq 0$ for all $t > 0$.

Definition 1.3 (**Pure Financing Project**).

A **pure financing project** is a project where $C_0 > 0$ and $C_t \leq 0$ for all $t > 0$.

Definition 1.4 (**Independent Projects**).

A and B are **independent projects** if the cash flows of A do not affect the cash flows of B.

Note. Between projects A and B, we can choose:

- one,
- both,
- or neither.

Definition 1.5 (**Mutually Exclusive Projects**).

A and B are **mutually exclusive projects** if choosing $A \implies$ we cannot choose B and vice versa (we cannot choose both).

Example. Given $C_0, C_1, C_2, \ldots$. How to decide if this is a "good" project?

Proof. See NPV rule below. $\square$

Definition 1.6 (**Net Present Value (NPV)**).

The **NPV** of a project is

$$NPV = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \cdots = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

where units = dollars.

Definition 1.7 (**NPV Rule**).

- For independent projects, choose all projects with $NPV > 0$.
- For mutually exclusive projects, choose the project with the highest NPV, as long as $NPV > 0$.

Example. Let $C_0 = -100, C_1 = 110, C_t = 0$ for $t \geq 2$, and $r = 8\%$. Then,

$$NPV = -100 + \frac{110}{1.08} > 0.$$

So, we should accept this project.

Example. Let $C_0 = 100, C_1 = -110$, and $r = 8\%$. Then,

$$NPV = 100 - \frac{110}{1.08} < 0.$$

So, we should reject this "bad" project.

What is r? $\implies$ r = "opportunity cost" of the dollar.

Advantages of NPV rule:

- NPV takes into account the time value of money.
- NPV takes into account cash flows, not accounting incomes.

Example. Let $C_0 = 100, C_1 = 115, r = 10\%$. The project has positive NPV and the owner will be happy if the project is taken.

Disadvantages of NPV rule:

- The cash flows are uncertain.
- What should be the right "r"? (see later: CAPM)

1.2 Review of ACTSC 231

Some important cash flows to recall from ACTSC 231:

- Propetuity.
- Propetuity with growth.
- Annuity.

Definition 1.8 (**Propetuity, Perpetuity Factor**).

A **propetuity** is an annuity with no end dates (infinite # of payments). Let $C_t = C$ for all $t \geq 1$. Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots \\ &= \frac{C}{1+r}\left[1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots\right] \\ &= \frac{C}{1+r} \cdot \frac{1}{1-\frac{1}{1+r}} \\ &= \frac{C}{r}. \end{aligned}$$

The **perpetuity factor** is $\frac{1}{r}$.

Definition 1.9 (**Perpetuity with Growth**).

Let $C_t = C(1+g)^{t-1}$ for $t \geq 1$. Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \cdots \\ &= \frac{C}{1+r}\left[1 + \frac{1+g}{1+r} + \left(\frac{1+g}{1+r}\right)^2 + \cdots\right] \\ &= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1+g}{1+r}} \\ &= \frac{C}{r-g}, \quad g < r. \end{aligned}$$

Definition 1.10 (**Annuity, Annuity Factor**).

Let $C_t = C$ for $1 \leq t \leq n$. Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C}{(1+r)^n} \\ &= \frac{C}{1+r}\left[1 + \frac{1}{1+r} + \cdots + \frac{1}{(1+r)^{n-1}}\right] \\ &= \frac{C}{1+r} \cdot \frac{1 - \left(\frac{1}{1+r}\right)^n}{1 - \frac{1}{1+r}} \\ &= C \cdot \frac{1-(1+r)^{-n}}{r}. \end{aligned}$$

The **annuity factor** is $\frac{1}{r}\left[1 - (\frac{1}{1+r})^n\right]$.

1.3 Other Investment Rules

Definition 1.11 (**Payback Period**).

The **payback period** is the # of years required to recoup the initial investment (ignoring the time value of money).

Example. Let $C_0 = -300, C_t = 100$ for $t \geq 1$. The payback period is 3 years.

Note. The units of payback period is years.

Definition 1.12 (**Payback Period Rule**).

Decide on a cutoff period, T^*. If the payback period $\leq T^*$, accept; otherwise, reject it.

Disadvantages of Payback Period Rule:

- It does not take into account the time value of money.
- The cutoff is arbitrary.
- Negative NPV may be accepted.
- Positive NPV may be rejected.

Definition 1.13 (**Discounted Payback Period**).

The **discounted payback period** is the payback period, but without ignoring the time value of money.

Example. Let $C_0 = -300, C_t = 100$ for $t \geq 1$, and $r = 12.5\%$. We are solving $\sum_{t=1}^{T} \frac{C_t}{(1+r)^t} = -C_0$. This implies that the discounted payback period, T, is 4 years.

Definition 1.14 (**Internal Rate of Return (IRR)**).

The **internal rate of return** (**IRR**) of a project is the cost of capital such that the NPV is zero.

Example. Let $C_0 = -100, C_1 = 112, r = 10\%$. We have two ways to determine if we should accept this project:

(1) Calculate NPV: $NPV = -100 + \frac{112}{1.1} > 0 \implies$ accept.

(2) The project has a return of 12%. Since it is bigger than $r = 10\%$, we should accept it.

Question: How to solve for the IRR?

Answer: Calculate the NPV and equal it to zero, then solve for r^*. That is,

$$\sum_{t=0}^{T} \frac{C_t}{(1+r^*)^t} = 0.$$

Definition 1.15 (IRR Rule).

Calculate the iRR of a project.

- For independent projects, choose all projects with IRR $> r$.
- For mutually exclusive projects, choose the project with the highest IRR (as long as IRR $> r$).

Advantages of IRR rule:

- In order to calculate IRR, we do not need r.
- It usually agrees with the NPV rule.

Disadvantages of IRR rule:

The first two disadvantages are for independent projects.

(1) IRR cannot distinguish between investment and financing.

Note. $(-1)\sum \frac{C_t}{(1+r^*)^t} = 0(-1)$, the solution remains the same.

Example. Let $C_0 = -100, C_1 = 112, r = 10\%$. The IRR is 12% and we should accept this project.

Let $C_0 = 100, C_1 = -112, r = 10\%$. The IRR is still 12% but we should reject this project.

(2) Polynomial equation problem: finding the IRR = solving a polynomial equation, i.e.

$$C_0 + \frac{C_1}{1+r^*} + \frac{C_2}{(1+r^*)^2} + \cdots + \frac{C_T}{(1+r^*)^T} = 0.$$

However, we could have no roots, multiple roots, imaginary roots, or complex roots.

Example. Let $C_0 = 0, C_1 = 100, C_t = 0$ for all $t \geq 2$. The IRR $= \infty$.

Example. Let $C_0 = -100, C_1 = 230, C_2 = -132$. Then,

$$-100 + \frac{230}{1+r^*} - \frac{132}{(1+r^*)^2} = 0 \implies r^* = 10\% \text{ or } 20\%.$$

This is problematic, since which IRR should we use?

We will need the following result to deal with the multiple IRR problem.

Theorem 1.1 (**Descartes' Rule of Signs**).

The # of positive roots of a polynomial is $\leq$ # of "sign switches" of the coefficients.

Example. For the previous example where $C_0 = -100, C_1 = 230, C_2 = -132$, there are 2 sign switches. Thus, there are at most 2 positive roots.

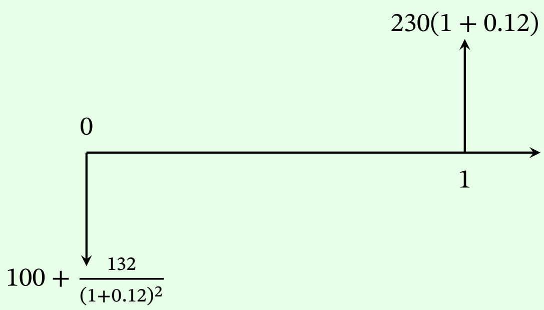
We have the following MIRR to avoid multiple solutions.

Definition 1.16 (**Modified Internal Rate of Return (MIRR)**).

The **modified IRR**

- brings all negative cash flows to time $t = 0$, and
- brings all positive cash flows to time $t = T$,

Example. Let $C_0 = -100, C_1 = 230, C_2 = -132$, and $r = 12\%$. Then,



Calculate the IRR of the modified project (MIRR):

$$-\left(100 + \frac{132}{(1+0.12)^2}\right) + \frac{230(1+0.12)}{1+r^*} = 0 \implies r^* = 12.36\%.$$

Note. We only have 1 "sign switch" in the modified project, so MIRR is unique. Then, we can compare MIRR with r, and choose the project with MIRR $> r$. But we need to know today's interest rate r.

Disadvantages of IRR rule (continued):

For mutually exclusive projects (i.e. we can only choose one project.):

(3) Scaling problem: the IRR is biased towards "smaller" projects.

Example.

Project A: $C_0 = -1, C_1 = 2$.

Project B: $C_0 = -100, C_1 = 150$.

Note that $\text{IRR}_A = 100\%$ and $\text{IRR}_B = 50\%$. However, project B is a better one.

Note. $\sum \frac{kC_t}{(1+r)^t} = 0$. IRR remains the same if you multiply by a constant.

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Here are some ways to deal with the scaling problem:

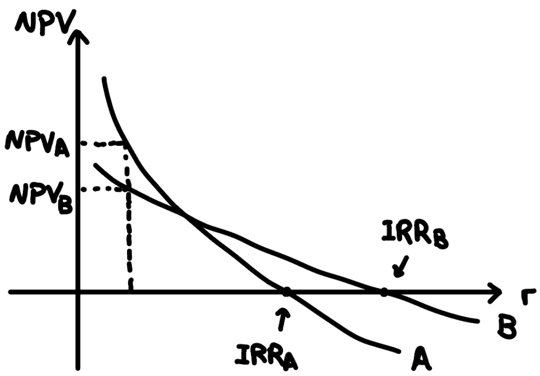
- Calculate the NPV and choose the higher one. In the example, $\text{IRR}_A > \text{IRR}_B$, but $\text{NPV}_A = 1 < \text{NPV}_B$.
- If the IRR picks the "smaller" project, then we do **incremental analysis**.

Example. Incremental project: $C = B - A$ with $C_0 = -99, C_1 = 148$.

If the IRR of the incremental project $< r \implies$ stick with the smaller project. If $> r \implies$ go with the bigger project.

(4) Timing problem.

Consider pure investment projects.



According to the IRR rule, B would be chosen, but A is better because $NPV_A > NPV_B$.

Example.

Project short: $C_0 = -300, C_1 = 345, C_t = 0$ for $t \geq 2$.

Project long: $C_0 = -200, C_4 = 330, C_t = 0$ for $t \neq 0, 4$.

Note that

$$IRR_{\text{short}} = 15\%, \quad NPV_{\text{short}} = 13.64$$
$$IRR_{\text{long}} = 13.34\%, \quad NPV_{\text{long}} = 25.39.$$

Note (**Recall from ACTSC 231: YTM of a Bond**).

A bond is a loan in exchange for regular coupon payments and the redemption value at maturity.

We have

$$P = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}$$
$$0 = -P + \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}.$$

Note that r is the interest rate that equates NPV to 0. So YTM = IRR of the bond.

Example. Let $C_0 = -100, C_1 = 10, C_2 = 110$ with $r = 10\%$. We can think of this as a bond with $P = 100, C = 10, F = 110, N = 2$. The IRR is 10%.

Also note that if the money was put in a bank, then we would have $C_0 = -100, C_2 = 121$. The extra \$1 is due to the \$10 invested at 10% for another year.

IRR assumes that all cash flows can be reinvested at the IRR itself.

A more realistic assumption that cash flows can be reinvested at r, not IRR.

Definition 1.17 (Profitability Index (PI)).

The **profitability index** is

$$PI = \frac{PV \text{ of future cash flows}}{\text{Initial Investment}} = \frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0}.$$

Definition 1.18 (**PI Rule**).

- For independent projects, choose all projects with $PI > 1$.
- For mutually exclusive projects, choose the project with the highest PI with $PI > 1$.

Proposition 1.2. For independent projects, PI agrees with the NPV rule.

Proof. Note that

$$\frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0} > 1 \implies NPV > 0.$$

□

Disadvantages of PI rule:

For mutually exclusive projects:

(1) Scaling problem (same as the IRR rule).

Example. Let $r = 10\%$.

Project A: $C_0 = -10, C_1 = 15, C_2 = 40$.

Project B: $C_0 = -20, C_1 = 70, C_2 = 10$.

Note that

$$PI_A = 4.53, \quad NPV_A = 35.3$$

$$PI_B = 3.52, \quad NPV_B = 50.5.$$

Note that $PI_A > PI_B$, but $NPV_A < NPV_B$. Let's consider the incremental project $C = B - A$ with $C_0 = -10, C_1 = 55, C_2 = -30$.

If the incremental project has $PI > 1$, then choose the bigger project; otherwise, choose the smaller project.

Summary of Investment Rules: NPV rule is the best.

1.4 Equivalent Annual Costs (EAC)

Example (**Machines with Unequal Lives**).

Let $r = 10\%$ and consider the following two machines.

A: \$500 to buy, \$120/year to operate, lasts for 3 years.

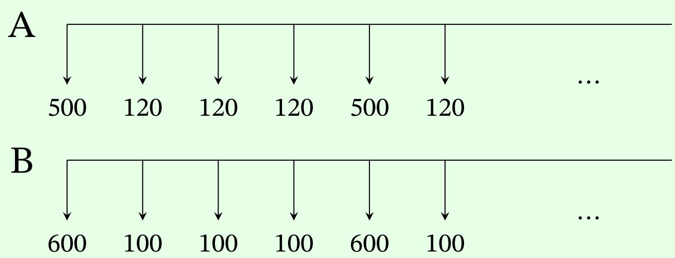
B: \$600 to buy, \$100/year to operate, lasts for 4 years.

Note that we have to use the machine (so we buy another one after it breaks down).

We can calculate the NPVs to get

$$NPV_A = -798.42, \quad NPV_B = -916.94.$$

However, the actual cash flows are different. We have the following cash flows:



Find the EAC of each machine.

Proof. This is equivalent to solving for x in $NPV_A = x \cdot a_{\overline{3}|10\%}$ AND $NPV_B = x \cdot a_{\overline{4}|10\%}$. Then, we get

$$EAC_A = \frac{NPV_A}{a_{\overline{3}|10\%}}, \quad EAC_B = \frac{NPV_B}{a_{\overline{4}|10\%}}.$$

We choose the one with lower EAC! □

Goal: Calculate NPV, cash flows, r, in the right way!

2 Capital Budgeting

2.1 Rules for Calculating NPVs

Remember the folllowing rules when calculating NPVs:

(1) Ignore sunk costs, i.e. costs we have to incur with/without the project.

Example. Thaler Pizza experiment.

(2) Include opportunity costs, i.e. cost of the next best alternative.

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(3) Be aware of externalities, i.e. effects on other projects. One project can affect cash flows from other projects.

Example.
- Positive externality: synergy.
- Negative externality: erosion.

(4) Be aware of the difference between "real" and "nominal".

Note.

- Nominal Interest (i): in dollars.
- Real interest rate (r): amount of extra goods you can buy.
- Rate of inflation (π): increase with prices.

Example. Suppose you have \$1 today and price of the good is \$1. If you put \$1 in the bank, it will become $1 + i$ dollars in one year. Price of the good in one year will be $1 + \pi$. So the # of goods you can buy next year is

$$\frac{1+i}{1+\pi} = 1 + r.$$

The # of extra units of the good you can buy is (Fisher's Law):

$$r = \frac{1+i}{1+\pi} - 1 = \frac{i-\pi}{1+\pi}.$$

We can also approximate r by

$$r \approx i - \pi.$$

Rule for Calculating Cash Flows

- Be consistent. If we are using real cash flows, use real interest rates, otherwise use nominal cash flows and nominal interest rates.

  Note. Nominal cash flows to real cash flows:

  $$\text{Divide by } (1+\pi)^t.$$

  Real cash flows to nominal cash flows:

  $$\text{Multiply by } (1+\pi)^t.$$

Example. Suppose all cash flows are nominal. Let $C_0 = -1000, C_1 = 600, C_2 = 650, i = 14\%, \pi = 5\%$. Calculate the NPV.

Proof.

Method 1: Use nominal.

$$NPV = -1000 + \frac{600}{1+0.14} + \frac{650}{(1+0.14)^2}.$$

Method 2: Find the real interest rate from Fisher's Law:

$$1 + r = \frac{1+i}{1+\pi}.$$

Let C_0' be the real cash flow at time 0, then

$$\begin{aligned} C_0' &= C_0 \\ C_1' &= \frac{C_1}{1+\pi} \\ C_2' &= \frac{C_2}{(1+\pi)^2}. \end{aligned}$$

Then,

$$NPV = C_0' + \frac{C_1'}{1+r} + \frac{C_2'}{(1+r)^2}.$$

We should get the same answer using both methods. □

Remark (**Why does this work?**).

Suppose all cash flows are nominal. Then,

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+i} + \frac{C_2}{(1+i)^2} + \cdots \\ &= C_0 + \frac{\frac{C_1}{1+\pi}}{\frac{1+i}{1+\pi}} + \frac{\frac{C_2}{(1+\pi)^2}}{\left(\frac{1+i}{1+\pi}\right)^2} + \cdots \\ &= C_0' + \frac{C_1'}{1+r} + \frac{C_2'}{(1+r)^2} + \cdots. \end{aligned}$$

(5) Be aware of depreciation for cash flows from accounting statements.

Example.

Revenue	Cost	Depreciation	Taxes
1500	700	600	34%

Depreciation is a tax-deductible expense! For the taxable income,

$$\begin{aligned} \text{Taxable Income} &= \text{Revenue} - \text{Cost} - \text{Depreciation} \\ &= 1500 - 700 - 600 = 200. \end{aligned}$$

Also,

$$\text{Tax paid} = 0.34 \times 200 = 68.$$

Then,

$$\text{Cash Flow} = \text{Revenue} - \text{Cost} - \text{Tax paid} = 1500 - 700 - 68 = 732.$$

General approach: Let the cash flow be CF_t. Then,

$$\begin{aligned} CF_t &= \text{Revenues} - \text{Costs} - \text{Taxes} \\ &= (R_t - \text{Cost}_t) - (R_t - \text{Cost}_t - D)T_c \\ &= (R_t - \text{Cost}_t)(1 - T_c) + DT_c. \end{aligned}$$

The term $(R_t - \text{Cost}_t)(1 - T_c)$ is called the **after tax operating cash flow (ATOCF)**. Note that the term DT_c is the tax break we get because we can substract depreciation. And DT_c is the **capical cost allowance tax shield (CCATS)**.

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \cdots \frac{C_t}{(1+r)^t} \\ &= C_0 + \frac{C_1}{1+r} + \cdots + \frac{(\text{ATOCF} + \text{CCATS})}{(1+r)^t} \\ &= C_0 + PV\text{ATOCF} + PV\text{CCATS}. \end{aligned}$$

Note that $PV(\text{ATOCF})$ is the PV without subtracting depreciation, and $PV(\text{CCATS})$ is the PV of all tax breaks.

Note. To see the above, we have

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \cdots + \frac{C_t}{(1+r)^t} + \cdots \\ &= C_0 + \frac{(\text{ATOCF})_1 + (\text{CCATS})_1}{1+r} + \cdots + \frac{(\text{ATOCF})_t + (\text{CCATS})_t}{(1+r)^t} + \cdots \\ &= C_0 + \sum_{t=1}^{\infty} \frac{(\text{ATOCF})_t}{(1+r)^t} + \sum_{t=1}^{\infty} \frac{(\text{CCATS})_t}{(1+r)^t}. \end{aligned}$$

Question: How is depreciation (D) calculated?

Example. Suppose $d = 20\%$. A machine costs \$1 million dollars.

Year	UCC Beginning	Depreciation (D)	UCC End
1	\$1,000,000	\$200,000	\$800,000
2	\$800,000	\$160,000	\$640,000
3	\$640,000	\$128,000	\$512,000

Note that $D = d \cdot (\text{UCC Beginning})$, where UCC is the **undepreciated capital cost**.

Half-Year Rule: if you buy an asset, only 50% of the purchase price can be added to the first year. The remaining 50% can be added in the second year.

Example (Half-Year Rule).

Suppose $d = 20\%$ and a machine costs \$1 million dollars.

Year	UCC Beginning	Depreciation (D)	UCC End
1	\$500,000	\$100,000	\$400,000
2	\$900,000	\$180,000	\$720,000
3	\$320,000	\$144,000	\$576,000

Theorem 2.1 (PVCCATS Formula).

We have

$$PV(\text{CCATS}) = \frac{C \cdot d \cdot T_c}{r + d} \cdot \frac{1 + 0.5r}{1 + r} - \frac{S \cdot d \cdot T_c}{r + d} \cdot \frac{1}{(1 + r)^n}$$

where

- C = cost of the asset.
- d = rate pf depreciation.
- n = lifetime of the asset.
- S = salvage value of the asset.

2.2 Capital Budgeting Under Uncertainty

Example. MATHSOC is thinking of installing a solar panel:

- Cost: $1 million.
- Annual operating cost: 10K.
- Lifetime: 15 years.
- Salvage: 100K.
- Revenues: $100/hour.
- # of sunny hours: 1500.
- $r = 10\%$.
- No taxes.

Find the NPV.

Proof. We have

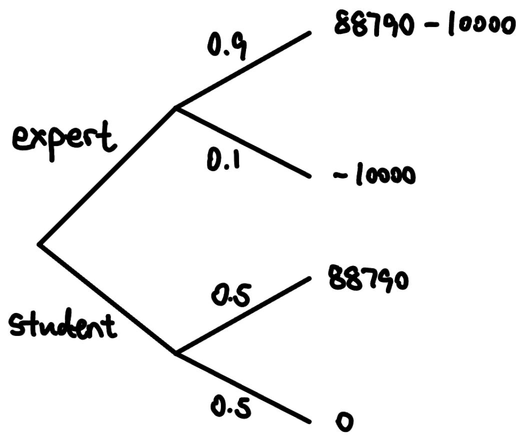
$$NPV = -1000000 - 10000a_{\overline{15}|10\%} + 150000a_{\overline{15}|10\%} + \frac{100000}{(1.1)^{15}} = 88790.$$

□

Decision Tree

For the MATHSOC example, installation may not work. Consider two options:

- Hire an expert: cost $10000, success rate 90%.
- Hire a student: cost $0, success rate 50%.



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There are two types of nodes in a decision tree:

- Probability nodes: calculate expected values.
- Decision nodes: choose the "better" path.

Example (Continued).

At the expert node, the expected value is $0.9(88790) + 0.1(-10000) = 76911$.

At the student node, the expected value is $0.5(88790) + 0.5(-10000) = 39395$.

Since $76911 > 39395$, we should hire the expert.

Scenario Analysis

For the MATHSOC example, what if the # of sunny hours $\neq 1500$? Suppose # of sunny hours are $1100, 1500, 1900$ with probability $\frac{1}{3}$ each.

We calculate the expected NPV, i.e. $\mathbb{E}[NPV] = \frac{1}{3}(NPV_{1100} + NPV_{1500} + NPV_{1900})$.

Monte Carlo Simulation

Suppose sunny hours $\sim$ N$(1500, 200)$. We simulate n observations from this distribution, and calculate NPVs for each outcome, and looke at the distribution of NPVs.

Sensitivity Analysis

How sensitive is our NPV to different variables? Let S = # of sunny hours. Then, we want to find $\frac{\mathrm{d}NPV}{\mathrm{d}S} \implies$ how accurate we need to be with our estimates.

Break-Even Analysis

Break-even point is the value of the variable that makes NPV $= 0$.

Definition 2.1 (Contribution Margin).

The **contribution margin** is the increase in net income per unit increase in sales, i.e.

$$\text{Contribution Margin} = (\text{Unit Revenue} - \text{Unit Cost}) \cdot (1 - T_c)$$

where T_c is the corporate tax rate.

Example. The Tim Hortons at SLC sells donuts. Assume no taxes.

- Donuts: \$1.20.
- Cost of donuts: \$0.70.
- Fixed cost: $\$100,000$.

Contribution Margin $= 1.20 - 0.70 = 0.50$. Then, the break-even point for # of donuts sold is $\frac{100000}{0.50} = 200000$, i.e. we need to sell 200000 donuts to cover the fixed cost.

Example. Same setting as above example, but with tax = 40%. Then,

$$\text{Contribution Margin} = (1.20 - 0.70)(1 - 0.4) = 0.30.$$

Then, the break-even point for # of donuts sold is $\frac{100000}{0.30} \approx 333333$.

Example. The IRR is the break-even interest rate!

Example. The discounted payback period is the break-even time!

For the next few lectures, we will focus on the following question.

Question: What is the right "r" for a project? (CAPM)

2.3 Utility Theory

Before we get there, let's consider the following question.

Question: How do rational human beings evaluate uncertain outcomes?

Definition 2.2 (**Lottery**).

A lottery X is a random variable whose outcomes are dollars.

Example. Consider the lotteries X and Y with the following PMFs:

X	100	25
$\mathbb{P}(X = x)$	0.5	0.5

Y	81	49
$\mathbb{P}(Y = y)$	0.5	0.5

One way to compare lotteries is to compare their expected values.

Example (ST. Petersburg Paradox).

Let X be the amount of money you win from the following lottery. A fair coin is tossed until it comes up heads. If the first head appears on the n-th toss, you win 2^n dollars.

X	Probability
2	$\frac{1}{2}$
4	$\frac{1}{4}$
8	$\frac{1}{8}$
16	$\frac{1}{16}$
$\vdots$	$\vdots$

The expected value of the lottery is

$$\mathbb{E}[X] = \sum x_i p_i = \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 + \frac{1}{8} \cdot 8 + \cdots = 1 + 1 + 1 + \cdots = \infty.$$

So you should be willing to pay any finite amount to play this game. However, in reality, most people are not willing to pay a lot of money to play this game.

John-von-Neumann: Expected Utility Theory

We use the preference relation $\succeq$ to denote "is at least as good as". That is,

$$x \succeq y \implies x \text{ is at least as good as } y.$$

Question: What assumptions should $\succeq$ follow for rational agents?

Assumptions

(1) Completeness: $\forall X, Y$, either $X \succeq Y$ or $Y \succeq X$ (or both).

(2) Transitivity: $\forall X, Y, Z$, if $X \succeq Y$ and $Y \succeq Z$, then $X \succeq Z$.

(3) Independence: $\forall X, Y, Z$ and $\forall p \in (0, 1)$, $X \succeq Y \iff pX + (1-p)Z \succeq pY + (1-p)Z$.

Theorem 2.2. If $\succeq$ satisfies the above three assumptions, then $\exists$ a function $u(\cdot)$ such that $\forall X, Y$,

$$X \succeq Y \iff \mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)].$$

Note. $u(\cdot)$ is called the **von-Neumann Morgenstern utility function**. It is the utility you get after consuming x dollars.

Example. Consider the following lotteries with $u(x) = \sqrt{x}$.

X	100	25
$\mathbb{P}(X = x)$	0.5	0.5

Y	81	49
$\mathbb{P}(Y = y)$	0.5	0.5

For lottery X,

$$\mathbb{E}\left[u(X)\right] = 0.5\sqrt{100} + 0.5\sqrt{25} = 5 + 2.5 = 7.5.$$

For lottery Y,

$$\mathbb{E}\left[u(Y)\right] = 0.5\sqrt{81} + 0.5\sqrt{49} = 4.5 + 3.5 = 8.$$

2.4 Risk Aversion

Definition 2.3 (Risk Averse).
A person is a **risk averse** if (for any lottery), they prefer the expected values for sure over the lottery itself.

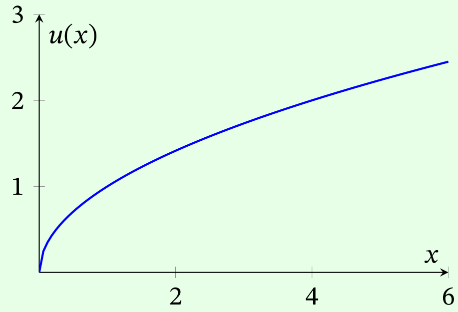
Definition 2.4 (Risk Neutral).
A person is **risk neutral** if they only care about the expected values.

Definition 2.5 (Risk Loving).
A person is **risk loving** if they prefer the lottery over the expected values.

Proposition 2.3. If the von-Neumann Morgenstern utility function $u(\cdot)$

- is concave, then the person is risk averse.
- is linear, then the person is risk neutral.
- is convex, then the person is risk loving.

Example. Let $u(x) = \sqrt{x}$. Note that $u''(x) < 0$, so the person is risk averse.



Decreasing marginal utility over money $\implies$ Risk averse, because the more money you have, the less utility you get from an extra dollar.

Say we have lotteries X and Y with the following PMFs:

X	100	0
$\mathbb{P}(X = x)$	0.5	0.5

Y	50
$\mathbb{P}(Y = y)$	1

Then,

$$\mathbb{E}[u(X)] = 0.5\sqrt{100} + 0.5\sqrt{0} = 5$$
$$\mathbb{E}[u(Y)] = 1 \cdot \sqrt{50} \approx 7.07.$$

If $u(x) = x$, then $\mathbb{E}[u(X)] = 0.5 \cdot 100 + 0.5 \cdot 0 = 50$ and $\mathbb{E}[u(Y)] = 1 \cdot 50 = 50$. So the person is risk neutral.

If $u(x) = x^2$, then $\mathbb{E}[u(X)] = 0.5 \cdot 100^2 + 0.5 \cdot 0^2 = 5000$ and $\mathbb{E}[u(Y)] = 1 \cdot 50^2 = 2500$. So the person is risk loving.

Lecture 6, 2025/09/22

Recall that a lottery X is a random variable with dollars as outcomes, where as x = outcome of X. The PMF of X is $f(x) = \mathbb{P}(X = x)$.

Let $U(X)$ be the utility you get from lottery X. Recall Theorem 2.2:

If a human being is rational, $\exists u(\cdot)$ over money such that

$$X \succeq Y \iff \mathbb{E}\left[u(X)\right] \geq \mathbb{E}\left[u(Y)\right]$$

where $U(X) = \mathbb{E}\left[u(X)\right]$.

Proposition 2.4.

(1) A person is risk averse if $u(\mathbb{E}[X]) > \mathbb{E}[u(X)]$ for all X.

(2) A person is risk neutral if $u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$ for all X.

(3) A person is risk loving if $u(\mathbb{E}[X]) < \mathbb{E}[u(X)]$ for all X.

Definition 2.6 (Certainty Equivalent).

The **certainty equivalent** of a lottery X is $CE(X)$ = dollar value (for sure) that makes you equally happy as the lottery, i.e.

$$u(CE(X)) = \mathbb{E}\left[u(X)\right].$$

Example. Let X be a lottery with the following PMF:

X	125	64	0
$\mathbb{P}(X = x)$	1/3	1/3	1/3

Let $u(x) = x^{\frac{1}{3}}$. Then,

$$\mathbb{E}\left[u(X)\right] = \frac{1}{3}125^{\frac{1}{3}} + \frac{1}{3}64^{\frac{1}{3}} + \frac{1}{3}0^{\frac{1}{3}} = 3.$$

Then, the certainty equivalent is a solution to $u(x) = 3$, i.e.

$$x^{\frac{1}{3}} = 3 \implies x = 27.$$

Proposition 2.5.

(1) Risk averse: $C\mathbb{E}[X] < \mathbb{E}[X]$.

(2) Risk neutral: $C\mathbb{E}[X] = \mathbb{E}[X]$.

(3) Risk loving: $C\mathbb{E}[X] > \mathbb{E}[X]$.

Degree of Risk Aversion

Note that u'' measures how risk averse a person is. We use the following measures.

Definition 2.7 (**Absolute Risk Aversion (ARA)**).

The **absolute risk aversion (ARA)** is

$$\mathrm{ARA} = -\frac{u''(x)}{u'(x)}.$$

This is also called the Arrow-Pratt measure of absolute risk aversion.

Definition 2.8 (**Relative Coefficient of Risk Aversion (RRA)**).

The **relative coefficient of risk aversion (RRA)** is

$$\mathrm{RRA} = -\frac{xu''(x)}{u'(x)} = x \cdot \mathrm{ARA}\,.$$

Example. Let $u(x) = -e^{-\alpha x}$. Calculate the ARA.

Proof. Note that

$$u'(x) = \alpha e^{-\alpha x}, \qquad u''(x) = -\alpha^2 e^{-\alpha x}.$$

Then,

$$\mathrm{ARA} = -\frac{u''(x)}{u'(x)} = -\frac{-\alpha^2 e^{-\alpha x}}{\alpha e^{-\alpha x}} = \alpha.$$

□

Example. Let $u(x) = x^\alpha$ for $0 < \alpha < 1$. Calculate the RRA.

Proof. Note that

$$u'(x) = \alpha x^{\alpha-1}, \qquad u''(x) = \alpha(\alpha-1)x^{\alpha-2}.$$

Then,

$$\mathrm{RRA} = -\frac{xu''(x)}{u'(x)} = -\frac{x\alpha(\alpha-1)x^{\alpha-2}}{\alpha x^{\alpha-1}} = 1-\alpha.$$

□

2.5 Mean-Variance Utility

This is a different approach to measure risk aversion.

Definition 2.9 (**Mean-Variance Utility**).
Let X be a lottery with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \operatorname{Var}(X)$. The **mean-variance utility** is

$$U(X) = \mu - \frac{A}{2}\sigma^2$$

where A is a known constant.

Note. All mean-variance utility are risk averse.

In fact, A is the **coefficient of risk aversion**. The higher A is, the more risk averse the person is.

Remark (**Summary of Two Approaches**).

(1) $U(X) = \mathbb{E}[u(X)]$.

(2) $U(X) = \mu - \frac{A}{2}\sigma^2$.

Question: Are these two approaches related?

Suppose we have a von-Neumann Morgenstern utility function

$$u(x) = x - \alpha x^2.$$

Then, $u'(x) = 1 - 2\alpha x$. Note that we want $u'(x) > 0 \implies x < \frac{1}{2\alpha}$, i.e. we are looking at "small" gambles. Note that a von-Neumann Morgenstern agent maximizes

$$\mathbb{E}[u(X)] = \mathbb{E}\left[X - \alpha X^2\right] = \mathbb{E}[X] - \alpha\mathbb{E}\left[X^2\right].$$

Note that

$$\operatorname{Var}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2 \implies \mathbb{E}\left[X^2\right] = \operatorname{Var}(X) + (\mathbb{E}[X])^2.$$

Then,

$$\mathbb{E}[u(X)] = \mu - \alpha(\sigma^2 + \mu^2) = \mu - \alpha\sigma^2 - \underbrace{\alpha\mu^2}_{\approx 0} \approx \mu - \alpha\sigma^2.$$

Example. Suppose the von-Neumann Morgenstern utility function is

$$u(x) = -e^{-\alpha x}.$$

Let $X \sim \mathrm{N}(\mu, \sigma^2)$. Then, this person is also a mean-variance utility maximizer.

Proof. We will do the proof in the next tutorial. □

We need to find μ and σ^2 of a lottery. We estimate μ by the following:

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} r_t,$$

where r_t is the return at time t.

3 Capital Asset Pricing Model (CAPM)

3.1 Risk and Return

Question: How are the r_t's calculated?

Let P_t be the price of an asset today. Then, P_{t+1} is the price of the asset in 1 year. Let D_t be the dividends you get this year. Then, the return is

$$r_t = \frac{(P_{t+1} + D_t) - P_t}{P_t} = \frac{P_{t+1} - P_t}{P_t} + \frac{D_t}{P_t}$$

where $\frac{P_{t+1}-P_t}{P_t}$ is the **price yield** and $\frac{D_t}{P_t}$ is the **dividend yield**.

Example (**Holding Period Return**).

Suppose the bank offers the following deal:

- 4% for year 1.
- 8% for year 2.
- 12% for year 3.

What is the average return?

Proof. Let the average be x. Then,

$$(1+x)^3 = (1+0.04)(1+0.08)(1+0.12) \implies x = \sqrt[3]{1.04 \times 1.08 \times 1.12} - 1.$$

Note that x is the geometric mean in this case (not arithmetic mean). □

Measure of Volatility

Definition 3.1 (**Variance and Standard Deviation of Returns**).

The variance of the returns is defined as

$$\sigma^2 = \frac{1}{n-1} \sum (r_t - \mu)^2,$$

Also, $\sigma = \sqrt{\sigma^2}$ is the standard deviation.

Other Measures

Let X and Y be the following two distributions of returns where X has a long left tail and Y has a long right tail. Here, Y is preferred by most investors even though both have the same variance.

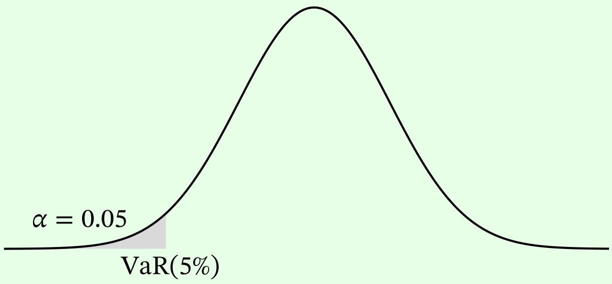
We need to come up with other measures of risk which capture the asymmetry.

Definition 3.2 (**Value At Risk (VaR)**).
VaR(α), where α is given as that value of x such that

$$\mathbb{P}(X \leq x) = \alpha.$$

In other words, VaR is the best case for the worst case scenarios.

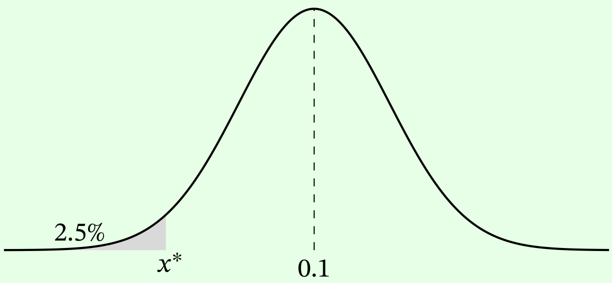
Example. VaR(5%) can be interpreted in the following plot.



Example. Let $X \sim \mathrm{N}(0.1, 0.08^2)$. Find VaR(2.5%).

Proof. Let x^* be such that $\mathbb{P}(X \leq x^*) = 2.5\%$. Then,

$$\mathbb{P}\left(\frac{X-\mu}{\sigma} \leq \frac{x^*-\mu}{\sigma}\right) = 2.5\% \implies \mathbb{P}(Z \leq y^*) = 0.025.$$



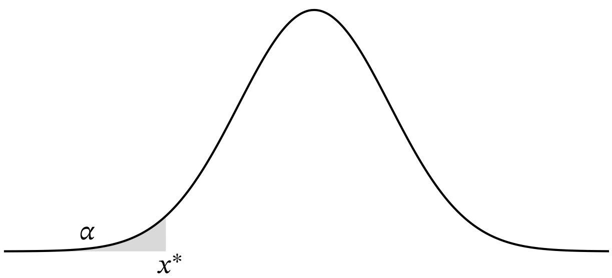
Thus, $y^* = \frac{x^* - \mu}{\sigma} = -1.96 \implies x^* = \mu + \sigma y^* = 0.1 + 0.08 \times (-1.96) = -0.0568$. Hence, $\text{VaR}(2.5\%) = -5.68\%$. □

Definition 3.3 (**Expected Loss (EL)**).

The **expected loss (EL)** or **Conditional Expected Loss (CEL)** is defined as

$$\text{EL}(\alpha) = \text{average return you make, conditional on the fact that } X \leq \text{VaR}(\alpha).$$

Look at the following plot.



Here, $x^* = \text{VaR}(\alpha)$ with $\mathbb{P}(X \leq x^*) = \alpha$. The CEL is the average of the shaded area. i.e.

$$\text{CEL}(\alpha) = \mathbb{E}\left[X \mid X \leq x^*\right].$$

Lecture 7, 2025/09/24

Recall some other measures of volatility:

- VaR.
- EL.
- Downside semi-variance.

Example. Suppose we have the following returns and probabilities.

X (%)	-15	-10	0	10	20
$f(x)$	0.02	0.03	0.5	0.4	0.05

Note that Var(5%) $= -10\%$. For EL(5%),

X (%)	-15	-10
Conditional probs	0.4	0.6

Then, EL(5%) $= -15\% \times 0.4 + (-10\%) \times 0.6 = -12\%$.

Definition 3.4 (**Downside Semi-Variance**).

The **downside semi-variance** is defined as

$$\mathrm{DSV} = \frac{1}{n-1} \sum_{r_i < \mu} (r_i - \mu)^2.$$

3.2 Capital Asset Pricing Model

Question: What portfolio will a risk-averse individual choose?

General Problem: Given n assets, a portfolio is $\mathbf{w} = (w_1, \ldots, w_n)$ where w_i is the proportion of wealth invested in asset i. Note that $\sum_{i=1}^n w_i = 1$. What should the w_i's be?

Capital Allocation Line

Suppose we are in a two-asset universe.

- Asset F: risk-free asset with return r_f.
- Asset P: risky asset with μ_p = average return, σ_p = volatility.

Example. Let $r_f = 7\%$, $\mu_p = 15\%$, $\sigma_p = 20\%$. A portfolio here is $(1-y, y)$, where y is the proportion invested in the risky asset and $1-y$ is the proportion invested in the risk-free asset. Suppose we invest \$10000. If $y = 0.3$, then we invest \$7000 in F and \$3000 in P.

The utility function is

$$U(x) = \mu - \frac{A}{2}\sigma^2.$$

Question: What should y be?

Example. Let $r_f = 7\%, \mu_p = 15\%, \sigma_p = 20\%$. Client 1 wants to invest with an average return of 10%. What should be y?

Proof. Note that

$$(1-y)r_f + y\mu_p = 10\% \implies y = \frac{10\% - 7\%}{15\% - 7\%} = 0.375.$$

□

Client 2 wants the volatility to be < 15%. What should be y?

Proof. Note that the portfolio of Client 2 is

$$\underbrace{C}_{Y} = \underbrace{(1-y)F}_{a} + \underbrace{y}_{b}\underbrace{P}_{X}.$$

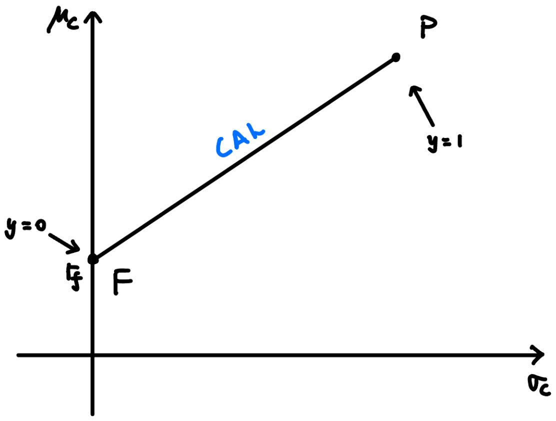
Remark. Recall from STAT 230, let $Y = a + bX$, then $\sigma_Y = |b|\sigma_X$.

Also, F is a constant and P is a random variable. Thus,

$$\sigma_C = y\sigma_P < 15\% \implies y < \frac{15\%}{20\%} = 0.75.$$

□

Question: What are the possible (μ, σ) pairs we can create?



Note that y = proportion invested in the risky asset P.

Question: What happens if y changes?

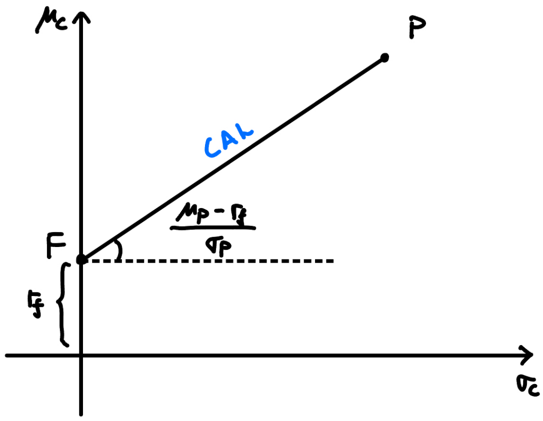
Proof. Note That

$$C = (1-y)F + yP$$
$$\mu_C = \mathbb{E}\left[(1-y)r_f + yP\right] = (1-y)r_f + y\mu_p = r_f + y\underbrace{(\mu_p - r_f)}_{\text{risk premium}}.$$

Also, $\sigma_C = y\sigma_p$. Then,

$$y = \frac{\sigma_C}{\sigma_P} \implies \mu_C = r_f + \frac{\sigma_C}{\sigma_P}(\mu_P - r_f) = r_f + \left(\frac{\mu_P - r_f}{\sigma_P}\right)\sigma_C.$$

$\square$



The slope of the CAL is $\frac{\mu_P - r_f}{\sigma_P}$, which is called the **Sharpe ratio** (or **reward-risk ratio**).

Definition 3.5 (Sharpe Ratio).

The ratio

$$\frac{\mu_P - r_f}{\sigma_P}$$

is called the **Sharpe ratio** or **reward-risk ratio** of the risky asset P.

Investor's Problem

We have the following optimization problem:

$$\max_C \mu_C - \frac{A}{2}\sigma_C^2 \text{ s.t. } C \text{ is on the CAL}.$$

To solve this,

$$\mu_C = r_f + y(\mu_P - r_f), \quad \sigma_C = y\sigma_P.$$

Then, the problem becomes

$$\max_y \{r_f + y(\mu_P - r_f)\} - \frac{A}{2}(y\sigma_P)^2.$$

Taking derivative and setting it to 0,

$$\mu_P - r_f - \frac{2Ay\sigma_P^2}{2} = 0 \implies y^* = \frac{\mu_P - r_f}{A\sigma_P^2}.$$

Note. This is why we have $\frac{1}{2}$ in the utility function $U(x) = \mu - \frac{A}{2}\sigma^2$.

Example. Let $f_f = 7\%$, $\mu_P = 15\%$, $\sigma_P = 20\%$, $A = 4$. What is y^*?

Proof. We have

$$y^* = \frac{\mu_P - r_f}{A\sigma_P^2} = \frac{15\% - 7\%}{4 \times (20\%)^2} = 0.5.$$

□

We saw the algebraic way of looking at this portfolio problem. Now, we look at it geometrically.

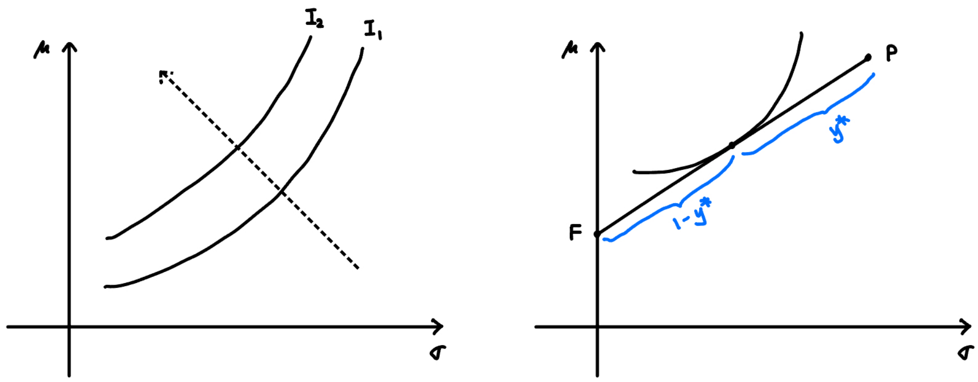
Geometric Approach

Definition 3.6 (Indifference Curve).
An **indifference curve** is defined as

$$\{(\sigma, \mu) : \text{ utility is the same for all points on the graph}\}.$$

That is, all points make you equally happy.

Note. Indifference curves have positive slopes.



Different indifference curves correspond to different levels of utility, and we want to move as top-left as possible. So maximizing happiness is equivalent to finding the tangent to the CAL (the most top-left indifference curve that touches the CAL).

Question: Can the indifference curves ever intersect?

Answer: No, because $A \sim B$ and $B \sim C$ implies $A \sim C$. That is, if A is as happy as B and B is as happy as C, then A is as happy as C.

Remark. $X \sim Y$ means X is indifferent to Y (i.e. they provide the same utility).

Question: Is it possible that $y^* > 1$?

Answer: Yes, we can borrow money and invest more in the risky asset.

Lecture 8, 2025/09/29

Question: For a risk-averse, mean-variance utility agent, what portfolio will they choose?

Case 1: One risk-free asset F and one risky asset P.

We have a portfolio $(1-y, y)$ where $y =$ proportion in the risky asset. For a porfolio C, we have

$$U = \mu_C - \frac{A}{2}\sigma_C^2.$$

Example. Let $r_f = 7\%$, $\mu_P = 15\%$, $\sigma_P = 20\%$. Suppose $y = 0.75$. What are μ_C and σ_C?

Proof. Note that

$$\mu_C = 0.25 \cdot 7\% + 0.75 \cdot 15\% = 13\%, \quad \sigma_C = 0.75 \cdot 20\% = 15\%. \qquad \square$$

The feasible set here is a straight line called the **Capital Allocation Line (CAL)**. The equation of the CAL is

$$\mu_C = r_f + \left(\frac{\mu_P - r_f}{\sigma_P}\right)\sigma_C$$

where the slope is the **Sharpe Ratio**.

We had the optimization problem:

$$\max_C \mu_C - \frac{A}{2}\sigma_C^2 \text{ s.t. } C \text{ is on the CAL} \implies y^* = \frac{\mu_P - r_f}{A\sigma_P^2}.$$

Note that y^* depends on the risk premium $\mu_P - r_f$, the volatility σ_P, and the risk aversion A.

Case 2: The two risky asset model.

Two Assets	Mean	SD
A	μ_A	σ_A
B	μ_B	σ_B

where $\sigma_A, \sigma_B > 0$. Let (w_A, w_B) be a portfolio, where w_A is the proportion invested in asset A and w_B is the proportion invested in asset B.

Question: What w_A, w_B should we choose?

For the feasible set, it is no longer linear (in general). We have

$$C = w_A A + w_B B, \quad w_A + w_B = 1.$$

Then,

$$\mu_C = w_A \mu_A + w_B \mu_B, \quad \sigma_C^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \operatorname{Cov}(A, B).$$

Remark (**STAT 230 Review**).

Recall that

$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X \mu_Y.$$

Let $X = \#$ of beers drunk, $Y =$ ACTSC 372 grades. If $x > \mu_X$, it is more likely that $y < \mu_Y$. Thus, $\operatorname{Cov}(X, Y) < 0$. The idea is that $\operatorname{Cov}(X, Y)$ is the average of the product of the deviations from the mean. It is also the average of the product minus the product of the averages.

Proposition 3.1 (**Properties of Covariance**).

(1) $\operatorname{Cov}(X, k) = 0$ for any constant k.

(2) $\operatorname{Cov}(X, X) = \operatorname{Var}(X)$.

(3) Bilinearity: $\operatorname{Cov}(aX + bY, Z) = a \operatorname{Cov}(X, Z) + b \operatorname{Cov}(Y, Z)$ for any constants a, b.

(4) If $X \perp\!\!\!\perp Y$, then $\operatorname{Cov}(X, Y) = 0$. (The converse is not true in general.)

Example (**Counterexample for Converse of (4)**). Let X be such that

X	-1	1
$f(x)$	$\frac{1}{2}$	$\frac{1}{2}$

Let $Y = X^2 = 1$ so that X and Y are not independent. Then,

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mu_X \mu_Y = 0 - 0 \cdot 1 = 0.$$

Note that Cov(X, Y) has units. We define the following to get a unitless measure.

Definition 3.7 (**Correlation Coefficient**).

The **correlation coefficient** between two random variables X and Y is

$$\rho_{XY} = \frac{\mathrm{Cov}\,(X, Y)}{\sigma_X \sigma_Y}.$$

Note that ρ_{XY} is unit independent.

Proposition 3.2 (**Properties of Correlation Coefficient**).

(1) $-1 \leq \rho_{XY} \leq 1$ for all X, Y.

(2) If $X \perp\!\!\!\perp Y$, then $\rho_{XY} = 0$.

(3) If $\rho = 0$, then there is no linear relationship between X and Y.

Recall that for any porfolio $C = (w_A, w_B)$, we have

$$\begin{aligned}
\mu_C &= w_A \mu_A + w_B \mu_B \\
\sigma_C^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \,\mathrm{Cov}\,(A, B) \\
&= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho_{AB} \sigma_A \sigma_B.
\end{aligned}$$

Special Case: $\rho = 1$

Note that

$$\begin{aligned}
\mu_C &= w_A \mu_A + w_B \mu_B \\
\sigma_C^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \sigma_A \sigma_B = (w_A \sigma_A + w_B \sigma_B)^2.
\end{aligned}$$

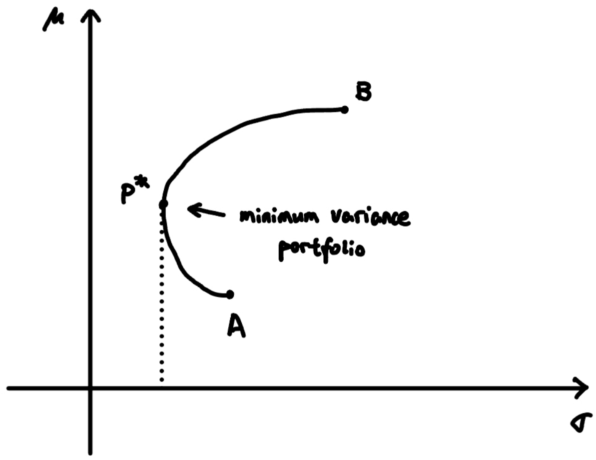
Then, $\sigma_C = w_A \sigma_A + w_B \sigma_B$. Note that σ_C is linear in w_A, w_B. Thus, the feasible set is a straight line.

If $\rho \neq 1$, then $\sigma_C < w_A \sigma_A + w_B \sigma_B$. The feasible set will be bowed towards the top left.

Value of Diversification

Diversification means to invest in multiple assets. This lowers σ for the same μ.

There are limits to diversification! Look at the plot below, p^* = minimum variance portfolio.



Special Case: $\rho = -1$

Note that

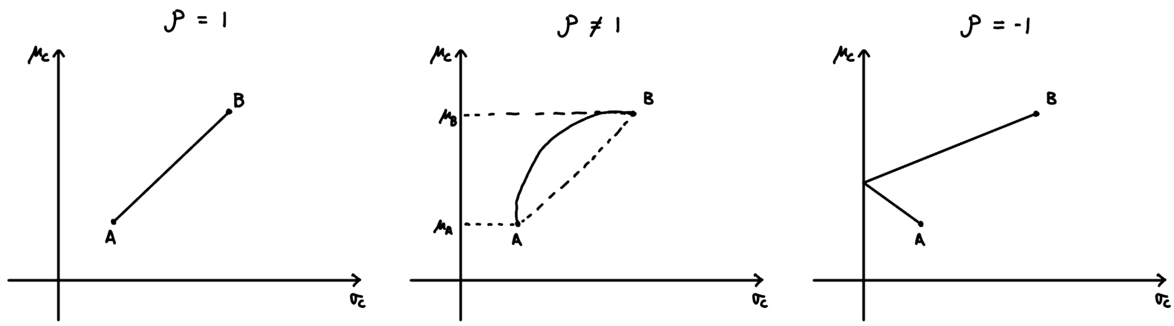
$$\begin{aligned}\sigma_C^2 &= w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B(-1)\sigma_A\sigma_B \\ &= (w_A\sigma_A - w_B\sigma_B)^2 \\ \implies \sigma_C &= |w_A\sigma_A - w_B\sigma_B|.\end{aligned}$$

Choosing

$$w_A = \frac{\sigma_B}{\sigma_A + \sigma_B}, \quad w_B = \frac{\sigma_A}{\sigma_A + \sigma_B}$$

gives $\sigma_C = 0$. All the risk can be diversified away (i.e. $\sigma_C = 0$). In general, it is <u>not true</u>.

Here is a summary of different ρ values:



Note. For $\rho \neq 1$, the curve is bowed towards the top left due to covariance term (lower σ_C for the same μ_C, compared to $\rho = 1$ case). For example, look at the midpoint of the line joining A and B.

Question: Can we find the minimum variance portfolio?

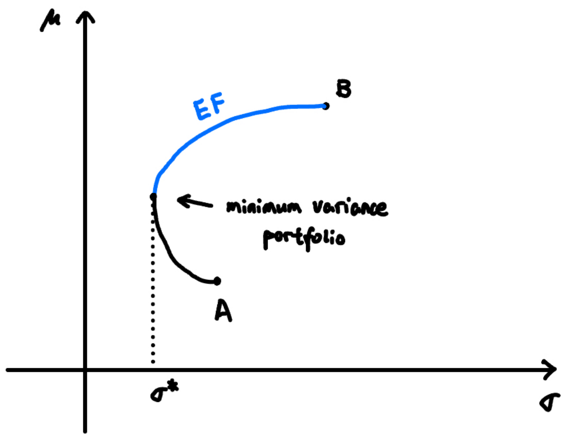
This is an optimization problem:

$$\min_{w_A, w_B} w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \rho_{AB}\sigma_A\sigma_B \text{ s.t. } w_A + w_B = 1.$$

Taking derivatives and setting to 0, we get

$$w_A^* = \frac{\sigma_B^2 - \rho_{AB}\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho_{AB}\sigma_A\sigma_B}, \quad w_B^* = 1 - w_A^*.$$

Definition 3.8 (**Efficient Frontier**).
The **Efficient Frontier** is the set of optimal portfolios, i.e. a portfolio is optimal if no other portfolio gives higher expected return AND lower risk at the same time.



All economic agents will be on the Efficient Frontier.

Example. Let $\mu_A = 8\%$, $\sigma_A = 12\%$, $\mu_B = 14\%$, $\sigma_B = 20\%$, $\rho = 0.3$. Let $w_A = 0.8, w_B = 0.2$. Then,

$$\mu_C = 0.8 \cdot 8\% + 0.2 \cdot 14\% = 9.2\%.$$

Also,

$$\sigma_C = \sqrt{0.8^2 \cdot 12\%^2 + 0.2^2 \cdot 20\%^2 + 2 \cdot 0.8 \cdot 0.2 \cdot 0.3 \cdot 12\% \cdot 20\%} = 11.4\%.$$

There are two sources of risk!

- Systematic risk.
- Idiosyncratic risk (firm-specific risk).

Lecture 9, 2025/10/01

Question: What should be the optimum portfolio for a risk averse agent?

Case 3: There are n risky assets, i.e. $A_1, A_2, \dots, A_n$. The mean vector is

$$\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)^\top.$$

The variance-covariance matrix is

$$\Sigma_{n\times n} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix}$$

where $\sigma_{ij} = \mathrm{Cov}\left(A_i, A_j\right)$ and $\sigma_{ii} = \mathrm{Var}\left(A_i\right) = \sigma_i^2$.

Proposition 3.3 (**Properties of** Σ**).**

(1) Σ is symmetric.

(2) There are no zero rows in Σ (i.e. no risk-free assets).

(3) Σ is invertible.

For $n = 2$, we have

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}.$$

Let $C = (w_1, w_2, \dots, w_n)$ be a portfolio such that $\sum_{i=1}^n w_i = 1$, i.e. $e^\top \mathbf{w} = 1$ where $e = (1, 1, \dots, 1)^\top$ and $\mathbf{w} = (w_1, w_2, \dots, w_n)^\top$.

Feasible Set

Question: What are the different μ and σ we can generate using these n assets?

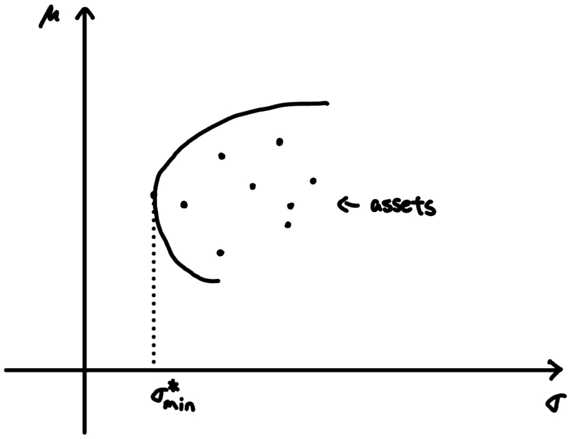
We have $C = w_1 A_1 + w_2 A_2 + \cdots + w_n A_n$. Then,

$$\begin{aligned}
\mu_C &= w_1\mu_1 + w_2\mu_2 + \cdots + w_n\mu_n = \mathbf{w}^\top \boldsymbol{\mu}, \\
\sigma_C^2 &= \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \operatorname{Cov}\left(A_i, A_j\right) \\
&= w_1^2\sigma_{11} + w_2^2\sigma_{22} + \cdots + w_n^2\sigma_{nn} + 2\sum_{i<j} w_i w_j \sigma_{ij} \\
&= \mathbf{w}^\top \Sigma \mathbf{w}.
\end{aligned}$$

Check variance for $n = 2$:

$$\sigma_C^2 = \begin{pmatrix} w_1 & w_2 \end{pmatrix}\begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = w_1^2\sigma_{11} + w_2^2\sigma_{22} + 2w_1w_2\sigma_{12}.$$

Question: What does the feasible set look like?



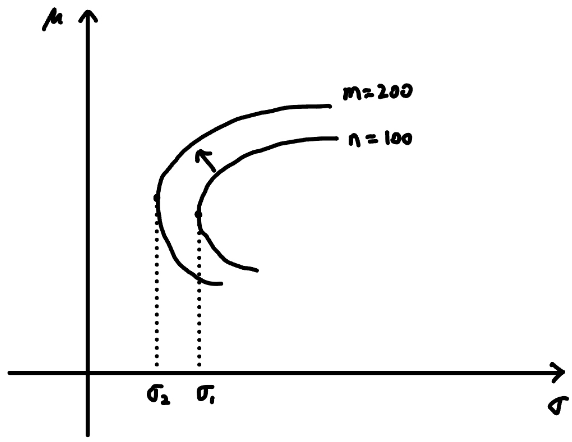
The feasible set is non-linear in general, and is bowed towards the top-left.

Remark (**Summary**).

- There is a value to diversification, i.e. σ_C^2 can be less than the weighted average of individual variances.
- There are limits to diversification.

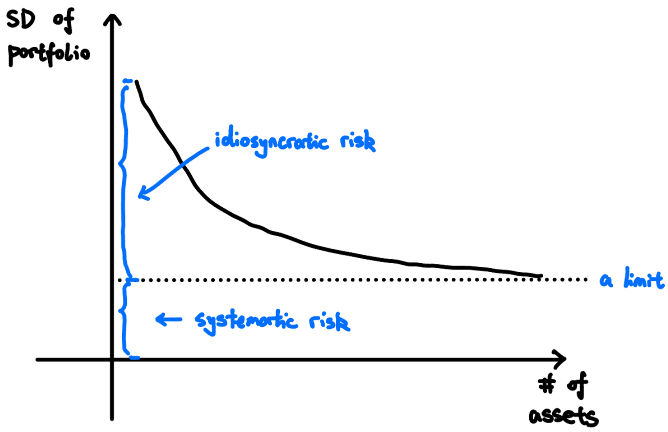
  It is possible that that all assets $A_1, \ldots, A_n$ lie below the feasible set.

As n increases, the feasible set moves towards the top-left:



There are two kinds of risks:

- Idiosyncratic risk (firm-specific risk).
- Systematic risk or market risk (affect all firms).



A well-diversified portfolio can get rid of all idiosyncratic risk, but not the systematic risk.

What matters are not the variances, but the covariances. That is, high-variance assets that have low covariances with other assets can reduce the overall portfolio risk, and vice versa.

Example. Suppose there are n assets each with the same standard deviation σ and the same correlation ρ (between any two assets). Consider the equally-weighted portfolio

$$w^\top = \left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right) = \text{equally-weighted portfolio.}$$

What is the variance of this portfolio?

Proof. We have

$$\begin{aligned}
\text{Var} &= \underbrace{\left(\frac{1}{n}\right)^2 \sigma^2 + \left(\frac{1}{n}\right)^2 \sigma^2 + \cdots + \left(\frac{1}{n}\right)^2 \sigma^2}_{n \text{ terms}} + n(n-1)\left(\frac{1}{n}\right)^2 \rho\sigma^2 \\
&= \frac{\sigma^2}{n} + \frac{n(n-1)}{n^2}\rho\sigma^2 \\
&= \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2.
\end{aligned}$$

□

Note that as $n \to \infty$, we have $\frac{\sigma^2}{n} \to 0$, whereas $\frac{n-1}{n}\rho\sigma^2 \not\to 0$. So, covariance effect will remain.

Some Technical Points

Here are some questions:

(1) Find the minimum variance portfolio for n-asset problem.

Note that for $(w_1, w_2, \dots, w_n)$, the optimization problem is

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \text{ s.t. } e^\top \mathbf{w} = 1.$$

We can use Lagrange multipliers to solve this and get

$$\mathbf{w}^* = \frac{\Sigma^{-1} e}{e^\top \Sigma^{-1} e}.$$

(2) How to find the Efficient Frontier (EF)?

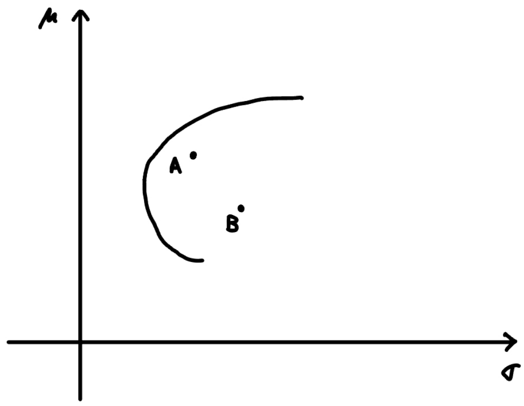
Method 1: Fix σ and maximize μ. That is,

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu} \text{ s.t. } \mathbf{w}^\top \Sigma \mathbf{w} = \sigma^2, \quad e^\top \mathbf{w} = 1.$$

Method 2: Fix μ and minimize σ. That is,

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \text{ s.t. } \mathbf{w}^\top \boldsymbol{\mu} = \mu, \quad e^\top \mathbf{w} = 1.$$

Even if asset A dominates asset B (i.e. $\mu_A > \mu_B$ and $\sigma_A < \sigma_B$), it is possible to put positive weights on B because of covariances.



Case 4: The $n + 1$ asset model.

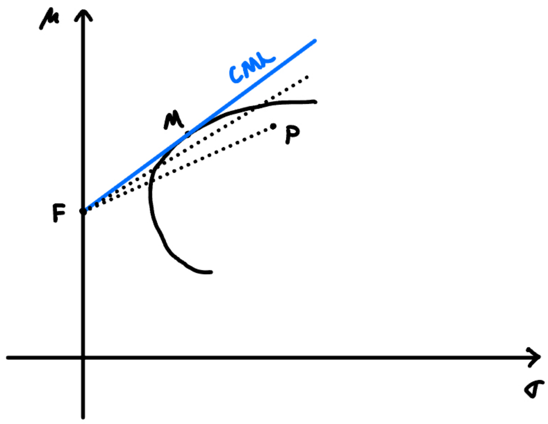
There are n risky assets $A_1, A_2, \ldots, A_n$ and one risk-free asset F.

Question: What should be the optimum portfolio?

Answer: First, solve the n risky asset problem, i.e. find the EF for the n risky assets.

Sub-Question: What risky asset should we choose?

Answer: The tangency portfolio M on the steepest CAL (i.e. CML).



All agents (irrespective of their risk aversion) will choose M as their optimum risky portfolio.

Definition 3.9 (**Capital Market Line (CML)**).

The **Capital Market Line (CML)** is the steepest CAL. It also maximizes the Sharpe ratio.

Note. CML is the new efficient frontier.

Question: What is the tangency portfolio M?

Example. Let $n = 2$ and let A and B be the two risky assets. Suppose $M = (0.2, 0.8)$, i.e. $w_A = 0.2$, $w_B = 0.8$. Then, the optimum risky portfolio is 20% in asset A and 80% in asset B.

Note. The tangency portfolio = Market share portfolio.

Example. Suppose there are 3 companies with the following market shares:

- A: \$5 million.
- B: \$3 million.
- C: \$2 million.

Then, $M = (0.5, 0.3, 0.2)$. The optimum risky portfolio is the market portfolio.

The optimum risky portfolio = market portfolio.

Mutual Fund Theorem states that all agents will choose the same risky portfolio M (i.e. the tangency portfolio/market portfolio) due to diversification.

ETF = Exchange Traded Fund, is a good approximation of M.

Passive investment is efficient (i.e. no need to pick stocks, just invest in the market portfolio).

Lecture 10, 2025/10/06

Recall:

(1) The trade-off between a risky asset and a riskless asset is linear (CAL). Agents will choose a point on the CAL (depending on A, the coefficient of risk aversion).

(2) The trade-off between risky assets (n-asset model) is non-linear, in general. The reason for this is the covariances.

Main points:

- Value of diversification.
- Limits to diversification (you cannot get rid of the systematic risk).

Efficiency frontier (EF). All agents (in the n-risky asset model) will be on the EF.

Note. To increase μ, you have to increase σ.

(3) The $n + 1$ asset model.

M = "best" risky portfolio = tangency portfolio.

CML = Capital Market Line. The slope of the CML is the Sharpe Ratio of M.

All agents will be on the CML.

Tangency Portfolio:

The tangency portfolio = market portfolio = $(w_1, \dots, w_n)$ where w_i = market share for firm i $= \frac{p_i x n_i}{\sum_j \mu_j n_j}$.

An ETF (Exchange Traded Fund) is a good approximation of M.

3.3 Security Market Line (SML)

Theorem 3.4 (**Capital Asset Pricing Model (CAPM)**).
Take any asset i in M. The CAPM equation is

$$\mu_i = r_f + \beta_i(\mu_M - r_f)$$

where

- μ_i = expected return of asset i (in percentage).
- r_f = fixed (risk-free) rate.
- μ_M = expected return of the market.
- $\beta_i = \frac{\mathrm{Cov}(A_i, M)}{\sigma_M^2}$ (measure of systematic risk of asset i).

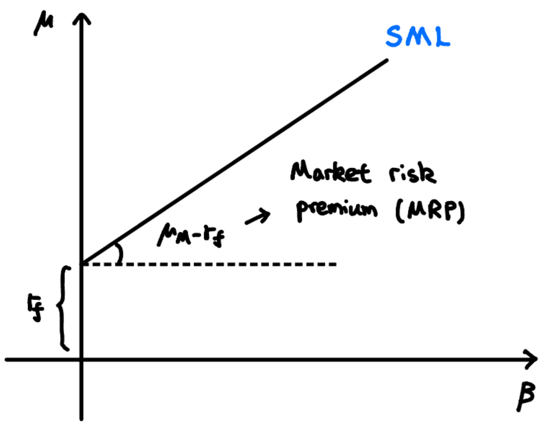
The "right" measure of risk is beta, whereas σ is the volatility.

Remark. The right interest rate for asset i is what you would have made on a similar asset (i.e. an asset with the same beta), which is given by the CAPM equation.

Definition 3.10 (**Security Market Line (SML)**).

The **Security Market Line (SML)** is the graph of the CAPM equation in the (β, μ)-plane:

$$\mu_i = r_f + \beta_i(\mu_M - r_f).$$



The SML gives us the expected return of an asset, given its beta. Note that

$$\beta_i = \frac{\mathrm{Cov}\,(A_i, M)}{\sigma_M^2} = \text{ how sensitive is asset } i \text{ to changes in the market.}$$

- $\beta = 1 \implies$ the asset is "as sensitive" as the market.
- $\beta > 1 \implies$ the asset is "more sensitive" than the market.
- $\beta < 1 \implies$ the asset is "less sensitive" than the market.

Example. Consider the project: $C_0 = -100$, $C_1 = 110$. Suppose $r_f = 2\%$, $\mu_M = 7\%$, $\beta = 1.1$. What is the "right" interest rate?

Proof. The averate return of a "similar" project is

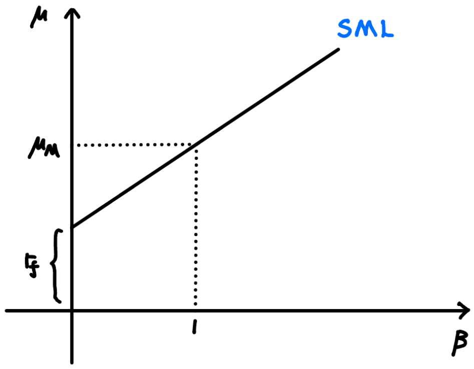
$$\mu_i = r_f + \beta_i(\mu_M - r_f) = 2\% + 1.1 \times (7\% - 2\%) = 7.5\%.$$

The interest rate for the project = expected value of a "similar" project (similar project means the same beta). $\square$

Question: Suppose $\beta = 1$. What should be the return?

Note that

$$\mu_i = r_f + 1 \cdot (\mu_M - r_f) = \mu_M.$$



If $\beta > 1 \implies \mu_i > \mu_M$ and if $\beta < 1 \implies \mu_i < \mu_M$.

If $\beta = 0$, then we have a risk-free asset. Note that

$$\mu_i = r_f + 0 \cdot (\mu_M - r_f) = r_f.$$

Question: Can we have $\beta < 0$?

Answer: Not sure.

Question: Suppose asset A has β_1 and asset B has β_2. Let $C = w_1 A + w_2 B$. What is β_C?

Proof. Given that

$$\beta_A = \frac{\mathrm{Cov}(A, M)}{\sigma_M^2}, \quad \beta_B = \frac{\mathrm{Cov}(B, M)}{\sigma_M^2}.$$

For C, we have

$$\begin{aligned} w_1\beta_A + w_2\beta_B &= w_1 \frac{\mathrm{Cov}(A, M)}{\sigma_M^2} + w_2 \frac{\mathrm{Cov}(B, M)}{\sigma_M^2} \\ &= \frac{w_1 \mathrm{Cov}(A, M) + w_2 \mathrm{Cov}(B, M)}{\sigma_M^2} \\ &= \frac{\mathrm{Cov}(w_1 A + w_2 B, M)}{\sigma_M^2} \\ &= \frac{\mathrm{Cov}(C, M)}{\sigma_M^2} = \beta_C. \end{aligned}$$

□

Example.

- Beta for food: low. So expected return is low.
- Beta for cryptocurrency: high. So expected return is high.

Question: How to find betas?

From the SML,

$$\mu_i = r_f + \beta_i(\mu_M - r_f)$$
$$\underbrace{(\mu_i - r_f)}_{y_i} = \beta_i \underbrace{(\mu_M - r_f)}_{x_i}$$

Collect (x_i, y_i) for $i = 1, 2, \dots, n$ and run a regression. Then,

$$\hat{\beta} = \text{ the estimate for } \beta.$$

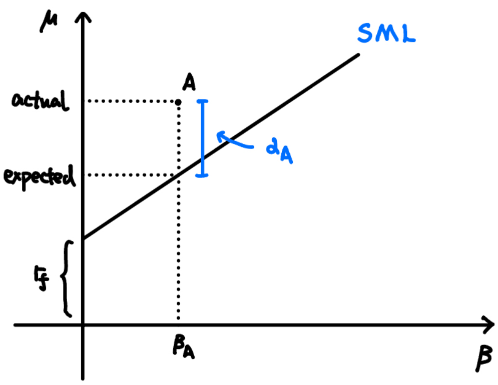
Performance Measures

Definition 3.11 (**Jensen's Alpha**).

The **Jensen's Alpha** of asset i is defined by

$$\alpha_i = \text{actual return of asset } i - \text{return predicted by SML}$$

- If $\alpha > 0$, then the asset is underpriced.
- If $\alpha < 0$, then the asset is overpriced.



Definition 3.12 (Traynor Ratio).

The **Traynor Ratio** of asset P is defined by

$$\mathrm{TR} = \frac{\mu_P - r_f}{\beta_P}.$$

If the asset lies on the SML, then $\mathrm{TR} = \mu_M - r_f$.

Example. XYZ is an all-equity firm with $\beta = 1.21$. Let $\mathrm{MRP} = 4\%$, $r_f = 6\%$, where MRP is the market risk premium, i.e. $\mu_M - r_f$. Let $C_0 = -100$ and

Projects	C_1
A	140
B	120
C	109

What projects should we choose?

Proof. The "right" interest rate is

$$\mu_i = r_f + \beta_i(\mu_M - r_f) = 6\% + 1.21 \times 4\% = 10.84\%.$$

So A and B are profitable projects, whereas C is not. □

Lecture 11, 2025/10/08

Recap:

The expected return of an asset depends on the following:

(1) r_f = risk-free rate.

(2) MRP = market risk premium = $\mu_M - r_f$.

(3) β_i = how sensitive your asset is to the market (appropriate measure of risk).

Note that MRP is the "common factor" for all assets and β_i is an appropriate measure of risk.

Proposition 3.5 (**Properties of Beta**).

(1) $\beta_i = 1 \implies \mu_i = \mu_M$.

(2) $\beta_i > 1 \implies \mu_i > \mu_M$.

(3) $\beta_i < 1 \implies \mu_i < \mu_M$.

(4) $\beta_i = 0 \implies \mu_i = r_f$.

Suppose asset A has β_1 and asset B has β_2. Let $C = w_1 A + w_2 B$. Then,

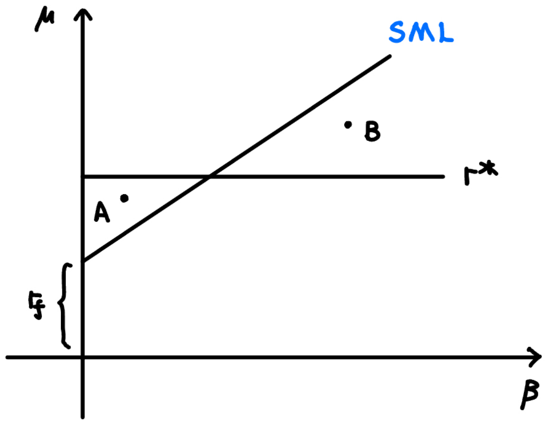
$$\beta_C = w_1 \beta_A + w_2 \beta_B.$$

Definition 3.13 (**Hurdle Rate**).

The **Hurdle Rate** is the minimum acceptable rate of return on an investment.

Note. That is, if the expected return of a project is $> r^*$, then accept the project; otherwise, reject.

Some companies have a fixed hurdle rate r^*, which can lead to bad investment decisions.



Note that project A is above the SML, so it is a good project. However, the hurdle rate rule suggests rejecting it. Also, project B is below the SML, but the hurdle rate rule suggests accepting it.

Example. Suppose a 2 risky asset model with $\rho = 0$ and the following data:

	# of shares	PPS	μ	σ
A	400	50	8%	12%
B	1000	80	14%	20%

(1) Find the minimum variance portfolio.

Proof. We have

$$w_A^* = \frac{\sigma_B^2 - \rho_{AB}\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho_{AB}\sigma_A\sigma_B} = \frac{20\%^2 - 0}{12\%^2 + 20\%^2 - 0} = \frac{400}{544} = 0.7353,$$
$$w_B^* = 1 - w_A^* = 0.2647.$$

So, the minimum variance portfolio is $\begin{pmatrix} 0.7353 \\ 0.2647 \end{pmatrix}$. □

Note that the optimization problem here is

$$\min_{w_A} w_A^2\sigma_A^2 + (1 - w_A)^2.$$

(2) What is M?

Proof. Company A's value is $400 \times 50 = 20000$. Company B's value is $1000 \times 80 = 80000$. So, A's market share is

$$w_A = \frac{20000}{20000 + 80000} = 0.2,$$

so the market portfolio is $M = (0.2, 0.8)$. □

(3) Find β_A.

Proof. We have

$$\beta_A = \frac{\operatorname{Cov}(A, M)}{\sigma_M^2}.$$

The numerator is

$$\operatorname{Cov}(A, 0.2A + 0.8B) = 0.2\operatorname{Cov}(A, A) + 0.8\operatorname{Cov}(A, B) = 0.2\sigma_A^2 + 0.8 \cdot 0.$$

Plug into the formula to get β_A. □

(4) Find β_B using an alternative method.

Proof. Note that $\beta_M = 1$ and $M = 0.2A + 0.8B$. So,

$$\beta_M = 0.2\beta_A + 0.8\beta_B = 1 \implies \beta_B = \frac{1 - 0.2\beta_A}{0.8}.$$

We can solve for β_B using part (3). $\square$

(5) Find r_f.

Proof. Create asset C such that $\beta_C = 0$:

$$C = w_A A + w_B B.$$

We must have the following:

$$\begin{cases} w_A + w_B = 1 \\ \beta_C = w_A\beta_A + w_B\beta_B = 0 \end{cases}$$

Solve for $\hat{w}_A, \hat{w}_B$, then

$$r_f = \mu_C = \hat{w}_A\mu_A + \hat{w}_B\mu_B.$$

$\square$

4 Arbritrage Pricing Theory (APT)

4.1 APT and Factor Models

From CAPM,

$$\mu = r_f + \beta \cdot \text{common factor}.$$

Instead of one common factor, we allow for k common factors.

- $k = 1 \implies$ single factor model.
- $k > 1 \implies$ multi-factor model.
- $k = 1$ and the common factor is the market: market model.

Assumptions of APT

A1 Individual assets assumption: there are n firms, the actual return of asset i is

$$R_i = \mu_i + m_i + \epsilon_i,$$

where

- R_i is the actual return of asset i.
- μ_i is the expected return of asset i.
- m_i is the set of common factors.
- ϵ_i is the firm-specific individual factor.

We have

$$m_i = \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k.$$

There are k common factors $F_1, F_2, \ldots, F_k$ with β_{ij} = sensitivity of ith firm to the factor j.

Example. The firm is Air Canada. The common factors are

$$m_i = \begin{cases} \text{Oil prices.} \\ \text{GDP of the world.} \\ \text{Inflation.} \end{cases}$$

And ϵ_i = pilots going on strike.

For $k = 1$,

$$R_i = \mu_i + \beta_i F + \epsilon_i.$$

We also assume the following:

- $\mathbb{E}\left[F_j\right] = 0 \; \forall j = 1, \dots, k$, where F is the **surprise factor**.
- $\mathbb{E}\left[\epsilon_i\right] = 0$.
- $\mathrm{Cov}\left(F_j, F_{j'}\right) = 0$ for $j \neq j'$ and $\mathrm{Cov}\left(F_j, \epsilon_i\right) = 0 \; \forall i, j$.

A2 Portfolio assumptions: a well-diversified portfolio has $\epsilon = 0$.

Example. For $k = 1$, a well-diversified portfolio has

$$\begin{cases} R_1 = \mu_1 + \beta_1 F + \epsilon_1, \\ R_2 = \mu_2 + \beta_2 F + \epsilon_2, \\ \vdots \\ R_n = \mu_n + \beta_n F + \epsilon_n. \end{cases}$$

Consider the portfolio C with weights $w_1, w_2, \dots, w_n$:

$$C = w_1 R_1 + w_2 R_2 + \cdots + w_n R_n.$$

Then,

$$R_c = \mu_c + \beta_c F + \underbrace{(w_1 \epsilon_1 + w_2 \epsilon_2 + \cdots + w_n \epsilon_n)}_{=0}$$

where $\mu_c = w_1 \mu_1 + w_2 \mu_2 + \cdots + w_n \mu_n$ and $\beta_c = w_1 \beta_1 + w_2 \beta_2 + \cdots + w_n \beta_n$.

A3 No arbitrage assumption: if two assets give you the same future returns, they must cost the same today.

Theorem 4.1. Under assumptions A1, A2, and A3, take any asset i,

$$\mu_i = r_f + \beta_{i1} \gamma_1 + \cdots + \beta_{ik} \gamma_k,$$

Note. $k = 1 \implies \mu_i = r_f + \beta_i \gamma$ (i.e. CAPM). Take $\gamma = M$, Then

$$\mu_i = r_f + \beta_i M.$$

Since $M = \mu_M - r_f$, we have $\mu_i = r_f + \beta_i(\mu_M - r_f)$.

Lecture 12, 2025/10/20

Recall:

- CAPM: $\mu_i = r_f + \beta_i \cdot \text{MRP} = \mu_M - r_f$, where MRP is the systematic factor.
- APT: instead of one systematic factor, we have multiple systematic factors, i.e. multi-factor model.
- APT Assumptions:
  - A1: Take any asset i,

    $$R_i = \mu_i + \underbrace{m_i + \epsilon_i}_{\text{noise}},$$

    Noise has two components:

    * $m_i = \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k$ where β_{ij} = sensitivity of the ith asset to the jth factor and F_j = common factor j.

      Example. Let F_j = oil prices, i = Ford, and i' = Tesla. Note that betas can act in the opposite direction for the same factor.

    * ϵ_i = idiosyncratic factor = firm-specific noise.

    We assume

    * $\mathbb{E}\left[F_j\right] = 0 \; \forall j$.
    * $\mathbb{E}\left[\epsilon_i\right] = 0 \; \forall i$.
    * $\text{Cov}\left(F_j, F_{j'}\right) = 0 \; \forall j \neq j'$ and $\text{Cov}\left(F_j, \epsilon_i\right) = 0 \; \forall i, j$.
  - A2: Take any well-diversified portfolio i. In such a portfolio, $\epsilon = 0$. Thus,

    $$R_i = \mu_i + m_i.$$

- A3: No arbitrage assumption. If two assets give you the same payoff, the price must be the same today.

Summary of A1 and A2:

- For individual assets: $R_i = \mu_i + \beta_{i1} F_1 + \beta_{i2} F_2 + \cdots + \beta_{ik} F_k + \epsilon_i$.
- For portfolios: $R_i = \mu_i + \beta_{i1} F_1 + \beta_{i2} F_2 + \cdots + \beta_{ik} F_k$.

Special Cases

- $k = 1$ (single factor model) for portfolios: $R_i = \mu_i + \beta_i F$.
- $k == 1$ and $F = M$ (market factor model) for portfolios: $R_i = \mu_i + \beta_i M$. Then, $\mathbb{E}[R_i] = \mu_i$ and $\mathrm{Var}(R_i) = \beta_i^2 \sigma_M^2$.

Recall Theorem 4.1: Under assumptions A1, A2, and A3, take any asset i,

$$\mu_i = r_f + \beta_{i1}\gamma_1 + \cdots + \beta_{ik}\gamma_k.$$

Special case: for $k = 1$, $\mu_i = r_f + \beta_i \gamma$.

Proof of Theorem 4.1.
We prove the case $k = 1$. Let us take two portfolios A and B. Then,

$$R_A = \mu_A + \beta_A F,$$
$$R_B = \mu_B + \beta_B F.$$

Question: Can we create an asset C such that C is risk-free?

We need to find weights w_A, w_B such that

$$w_A + w_B = 1, \quad w_A \beta_A + w_B \beta_B = 0.$$

Solving this, we have

$$w_A = \frac{\beta_B}{\beta_B - \beta_A}, \quad w_B = \frac{-\beta_A}{\beta_B - \beta_A}.$$

By no-arbitrage assumption, this asset C should give us $\mu_C = r_f$. So,

$$r_f = w_A \mu_A + w_B \mu_B.$$

Then,

$$r_f = \frac{\beta_B}{\beta_B - \beta_A}\mu_A + \frac{-\beta_A}{\beta_B - \beta_A}\mu_B \implies \frac{\mu_A - r_f}{\beta_A} = \frac{\mu_B - r_f}{\beta_B} = \gamma.$$

Thus,

$$\mu_i = r_f + \beta_i \gamma \quad \text{for } i = A, B.$$

□

Market Model

We have

$$\mu_i = r_f + \beta_i M$$

for all portfolios. Here,

$$\mu_M = r_f + \beta_M M, \quad M = \mu_M - r_f.$$

So, $\mu_i = r_f + \beta_i(\mu_M - r_f)$.

Example. Suppose we have a two factor model. We have

$$R_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \epsilon_i.$$

Let A, B, and C be three well-diversified portfolios with the following.

Portfolio i	β_{i1}	β_{i2}	μ_i
A	1.0	1.5	10%
B	0.5	1.0	7%
C	0.75	1.25	9%

Can we create a portfolio D using just A and B such that D is independent of factor 1?

Proof. We find weights w_A, w_B such that

$$w_A + w_B = 1, \quad w_A\beta_{A1} + w_B\beta_{B1} = 0.$$

□

What is the risk-free rate implied in this model?

Proof. Let w_A, w_B, w_C be the weights for a risk-free portfolio. We have

$$\begin{cases} w_A + w_B + w_C = 1, \\ 1.0w_A + 0.5w_B + 0.75w_C = 0, \\ 1.5w_A + 1.0w_B + 1.25w_C = 0. \end{cases}$$

We can solve for w_A, w_B, w_C. Then, the return of this portfolio (i.e. the risk-free rate) is

$$w_A\mu_A + w_B\mu_B + w_C\mu_C = r_f.$$

□

Note. APT is more practical than CAPM since the number of variables that need to be estimated under APT is less than that under CAPM.

4.2 Factors Affecting Beta

We have the following factors affecting beta:

- Cyclicality of revenues (how sensitive stocks are to business cycles).
- Operating leverage.
- Financial leverage.

Cyclicality of Revenues

An asset that is more sensitive to business cyclese will have a higher beta.

Example.

- Groceries: low beta.
- Restaurants: higher beta.
- Luxury goods: high beta.

Operating Leverage

Definition 4.1 (Operating Leverage).
Operating leverage is defined by

$$\text{OL} = \frac{\%\text{ change in earnings}}{\%\text{ change in sales}}.$$

Higher operating leverage $\implies$ higher beta.

Example. The following information for a project is given.

- Revenue: \$250 per year forever.
- Cost: \$50 per year forever.
- Assume CAPM holds and $r_f = 5\%$, $\mathbb{E}[r_M] = 10\%$, $\beta = 0.6$.

What is the NPV of the project?

Proof. We find the "right" interest rate first.

$$\mu = r_f + \beta(\mathbb{E}[r_M] - r_f) = 0.05 + 0.6(0.10 - 0.05) = 0.08.$$

Then,

$$NPV = \frac{250}{0.08} - \frac{50}{0.08} = \$2500.$$

□

Suppose the cost was fixed. Then, what is the NPV of the project?

Proof. The "right" interest rate for revenues is 8%. The "right" interest rate for costs is $r_f = 5\%$.

Then,

$$NPV = \frac{250}{0.08} - \frac{50}{0.05} = \$2125.$$

□

Case 1: The project is less sensitive to sales.

Case 2: The project is more sensitive to sales.

Example (**Continued**).

What is the beta of the project where the cost is fixed?

Proof. Recall that

$$NPV = NPV_{\text{revenues}} + NPV_{\text{costs}} = \frac{250}{0.08} - \frac{50}{0.05}.$$

Then,

$$w_{\text{revenues}} = \frac{3125}{2125} = 1.47, \quad w_{\text{costs}} = \frac{-1000}{2125} = -0.47.$$

Thus,

$$\beta_{\text{project}} = 1.47 \cdot 0.6 + (-0.47) \cdot 0 = 0.88.$$ □

Financial Leverage

So far, we assumed all equity firms.

Question: What is the right interest rate if a firm has debt?

Lecture 13, 2025/10/22

Information Asymmetry

In Finance, this is called agency problems.

Example. In a second-hand car, suppose 50% of cars are lemons (bad cars) and 50% are good cars. A lemon is worth \$1000 and a good car is worth \$3000. Then, the buyer should only pay for \$1000 and accept the fact that they might get a bad car.

Example. Suppose a firm has a cash flow of $\$C$ in perpetuity. So,

$$NPV = \frac{C}{r}$$

where r is the cost of capital.

Liquidity

Higher liquidity $\implies$ easier to trade $\implies$ lower cost of capital.

Definition 4.2 (**Bid-Ask Spread**).
The **bid-ask spread** is defined by

$$\text{Bid-Ask Spread} = \text{Buying price} - \text{Selling price}.$$

Note. A bid-ask spread is higher if there is information asymmetry.

Financial Leverage

So far, we assumed that all firms are all equity. Now, we will allow for debt.

Question: What is the right interest rate if a firm has debt?

Properties of Equity Contracts

- Ownership contract.
- Residual claim.
- Limited liability.

Properties of Debt Contracts (Loan Contracts)

- Interest you pay is tax-deductible.
- Failure to pay interest may lead to bankruptcy.

Remark. Value of firm $V = D + E$ where D = market value of debt, E = market value of equity.

Note. Levered firms: $D > 0$ and unlevered firms: $D = 0$.

Questions:

(1) What is the optimum debt-equity ratio?

Answer: We will answer this later (Modigliani-Miller Theorems)

(2) Given. the debt-equity ratio, what is the "right" interest rate?

Answer: Weighted Average Cost of Capital (WACC).

4.3 Weighted Average Cost of Capital (WACC)

Case 1: No taxes. The WACC is

$$r_{\text{WACC}} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D$$

where we get r_E from the SML, and r_D is the interest rate on loans (debt).

Case 2: With taxes. The WACC is

$$r_{\text{WACC}} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D (1 - T_c),$$

where T_c is the tax rate.

Example. Suppose $D = \$1$ million, $r_D = 8\%$, amd tax rate $= 40\%$. Then,

$$r_D(1 - T_c) = 8\% \cdot (1 - 0.4) = 4.8\%.$$

Interest paid is $\$80,000$ and Tax break $= 0.4 \cdot 80,000 = \$32,000$. So,

$$\text{Effective interest} = 80,000 - 32,000 = 48,000.$$

Note. Interest is tax-deductible.

Example. A firm has the following information.

- Debt-equity ratio is 0.6.
- Cost of debt is 15.15% (i.e. r_D).
- Cost of equity is 20% (i.e. r_E).
- Tax rate is 34% (i.e. T_c).

Is this a good project?

Proof. The WACC is

$$r_{\text{WACC}} = \frac{D}{D+E} r_D (1 - T_c) + \frac{E}{D+E} r_E.$$

Note that $D = 0.6E$. Then, we can compute to get $r_{\text{WACC}} = 16.25\%$. Then, plug r_{WACC} (as the interest rate) into NPV formula to see if the project is good. □

Example. XYZ is a firm with the following information.

- Common shares: # of shares = 8.5 million, PPS = \$34, $\beta = 1.2$.
- Bonds: 200000 outstanding bonds ($F = 1000$), 15 year bonds, 7.5% semi-annual coupons are selling for 93% of par.
- Tax rate = 35%, MRP = 7%, $r_f = 5\%$.

Find r_{WACC}.

Proof. We have

$$r_{\text{WACC}} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D (1 - T_c).$$

Note that

$$E = \text{\# of shares} \times \text{PPS} = 8.5 \text{ million} \times 34 = 289 \text{ million}.$$

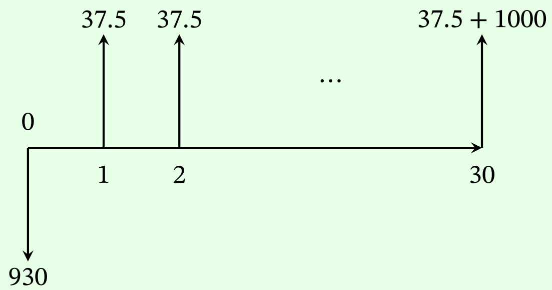
And,

$$D = \text{number of bonds} \times \text{price per bond} = 200,000 \times 930 = 186 \text{ million}.$$

To find r_E, use CAPM:

$$r_E = r_f + \beta \cdot MRP = 5\% + 1.2 \cdot 7\% = 13.4\%.$$

To find r_D, the cash flow of the bond is



Then, equate the NPV to 0 and solve for r_D:

$$0 = -930 + \sum_{t=1}^{30} \frac{37.5}{(1+r_D)^t} + \frac{1000}{(1+r_D)^{30}}.$$

Thus, we can compute to get $r_{\text{WACC}} = 10.27\%$. □

A Tutorials

A.1 Tutorial 1 (2025/09/19)

Mobius 2 Hints

Q1: Ignore CCA = cannot subtract depreciation.

Q10: $\sigma(NPV)$ is the standard deviation of NPV.

Note. For $Y = a + bX$, $\sigma_Y = |b|\sigma_X$.

Expected Utility

$U(X)$ = utility you get from lottery X. Under certain assumptions,

$$U(X) = \mathbb{E}\left[u(X)\right] = \sum u(x)f(x)$$

where $u(x)$ is the von-Neumann Morgenstern utility function and $f(x)$ is the PMF of X.

Risk Aversion

- A person is risk averse if $u(\mathbb{E}[X]) > \mathbb{E}[u(X)]$ for all X.
- A person is risk neutral if $u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$ for all X.
- A person is risk loving if $u(\mathbb{E}[X]) < \mathbb{E}[u(X)]$ for all X.

Note. $u(\cdot)$ is a concave function $\iff$ risk averse.

Example. Let X be a lottery with PMF $\mathbb{P}(X = 64) = \mathbb{P}(X = 8) = 0.5$ and let Y be a lottery with PMF $\mathbb{P}(Y = 125) = \mathbb{P}(Y = 0) = 0.5$. Consider $u(x) = x^{1/3}$ and $u(x) = \sqrt{x}$.
Note that for X with $u(x) = x^{1/3}$,

$$U(X) = 0.5 \cdot 64^{1/3} + 0.5 \cdot 8^{1/3} = 3.$$

For Y with $u(x) = x^{1/3}$,

$$U(Y) = 0.5 \cdot 125^{1/3} = 2.5.$$

Find $U(X)$ and $U(Y)$ for $u(x) = \sqrt{x}$.

Certainty Equivalent

Let $C\mathbb{E}[X]$ = amount of dollars for sure that makes you indifferent with the lottery X.

- Risk averse: $C\mathbb{E}[X] < \mathbb{E}[X]$.
- Risk neutral: $C\mathbb{E}[X] = \mathbb{E}[X]$.
- Risk loving: $C\mathbb{E}[X] > \mathbb{E}[X]$.

Definition A.1 (Certainty Equivalent).

The **certainty equivalent** of a lottery X is the amount of money $CE(X)$ such that

$$u(CE(X)) = \mathbb{E}[u(X)].$$

If $u(\cdot)$ is strictly increasing, $CE(X) = u^{-1}(\mathbb{E}[u(X)])$.

Example. Consider the following:

- Wealth: \$100.
- Fine: \$36 with probability $p = \frac{1}{4}$.
- M: \$9.25, parking fee to avoid the fine.
- Utility function: $u(x) = \sqrt{x}$.

Proof. Pay for parking $= \sqrt{100 - M}$. Don't pay for parking $= \frac{1}{4}\sqrt{100 - 36} + \frac{3}{4}\sqrt{100} = 9.5$. So $\sqrt{100 - M} > 9.5 \implies M < 100 - 9.5^2 = 9.75$. So we should pay for parking. □

Are We Rational?

Example (Allais Paradox).

Consider the following choices:

- A: \$1 million dollars with $p = 1$.
- B: \$1 million dollars with $p = 0.89$, \$5 million dollars with $p = 0.1$, and \$0 with $p = 0.01$.
- C: \$0 with $p = 0.9$, \$1 million dollars with $p = 0.1$.
- D: \$0 with $p = 0.89$, \$1 million dollars with $p = 0.11$.

If you prefer A over B and C over D, you are not rational.

A.2 Tutorial 2 (2025/10/03)